

QARTOD Status, Best Practices



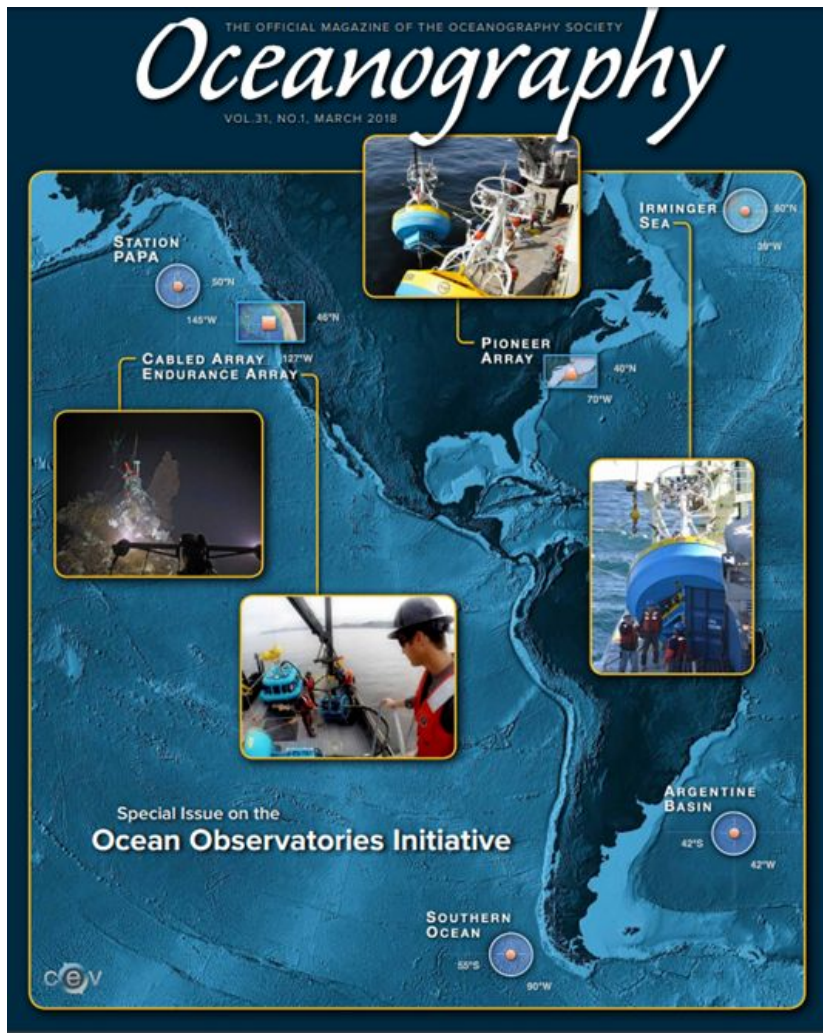
Mark Bushnell, U.S. IOOS
2018 IOOS DMAC Annual Meeting
May 21, 2018



QARTOD Manuals, Manual Maintenance



Who else is using QARTOD? 1



NSF Ocean Observatories Initiative

“Data Quality Control Processes - Oceanographic and engineering data throughout the OOI system are reviewed through manual (human in the loop) and automated quality control procedures. The overall goal is to ensure that the data and metadata delivered by the OOI meet community data quality standards. **These standards were designed with the goal of meeting the Integrated Ocean Observing System (IOOS) Quality Assurance of Real Time Ocean Data (QARTOD) standards.**”

Smith, L.M., J.A. Barth, D.S. Kelley, A. Plueddemann, I. Rodero, G.A. Ulses, M.F. Vardaro, and R. Weller. 2018. *The Ocean Observatories Initiative*. *Oceanography* 31(1):16–35, <https://doi.org/10.5670/oceanog.2018.105>.

After nearly **10 years** and **expenses of US\$386 million**, in June 2016, NSF announced that most OOI data were flowing in real time from more than **900 sensors at the 7 sites**. The **annual budget is approximately \$55 million**. Witze, Alexandra (2016-06-09). "US ocean-observing project launches at last". *Nature*. 534 (7606): 159–160. Bibcode:2016Natur.534..159W. doi:10.1038/534159a

U.S. Private Sector & University Classroom

- Jay Titlow / **Weatherflow** – Using wind tests for QC for their Caribbean installations
- Jeff Hansen / **WaveForce Technologies** - “We’ve applied QARTOD rules when rebuilding the USACE/FRF database.”
- Bruce Magnell / **Woods Hole Group** – “The QA/QC procedure implemented by WHG, in part follows QARTOD recommendations.”
- **Rutgers University** - Masters of Integrated Ocean Observing, a software/QA/QC boot camp informed by QARTOD.

International Use, Government & Private Sector

- British Oceanographic Data Centre - Global Sea Level Observing System (**GLOSS**) Quality Control Manual
- The **Southern Ocean Time Series** (SOTS) - Quality Assessment and Control Report
- **OMC International**, Australia - QC optimised for operational under keel clearance management purposes
- Mark Calverley / **Fugro UK** – “We've been advocating QARTOD in the oil and gas sector for quite a few years...”
- Carlos Garcia /SIMCosta – Implementing QARTOD in SiMCosta, the **Brazilian Coastal Monitoring System**.
- Christian Senet / **Bundesamt fuer Seeschifffahrt und Hydrographie** - “Have started to implement QARTOD wave QC testing.”

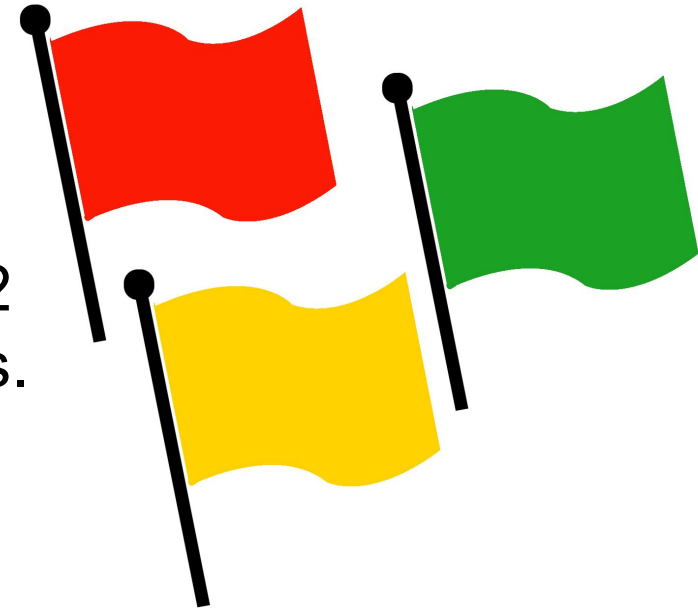
FY 2019 Tentative QARTOD Plans

- One new manual, perhaps pH
- Update two manuals
- Shift a bit toward QA, measurement uncertainty examples
- Continued international interaction
- Support for implementation, i.e. OMAO use aboard NOAA vessels, data flagging standards, others?

QC Data Flagging Implementation

Known

- Every real-time observation distributed to the ocean community must be accompanied by a quality descriptor.
- QARTOD (and IOC) suggest a Tier 2 flag for more detailed QC test results.



Unknown

- How standardized should RAs strive to be?
- How to meet NCEI data input constraints?
- How to meet User desires for various levels of QC?

QC Data Flagging Implementation (cont.)

Known issues:

Data , data flag

or

Data

Data flag

- Challenges (from Axiom, January webinar):
 - Transmitting quality flags around
 - Interface for displaying and communicating quality checks
 - Duplication of effort surrounding check implementations
 - Are flags served alongside the data? SOS/ERDDAP? NCEI?
 - Setting thresholds for third party stations
 - Documenting check configuration

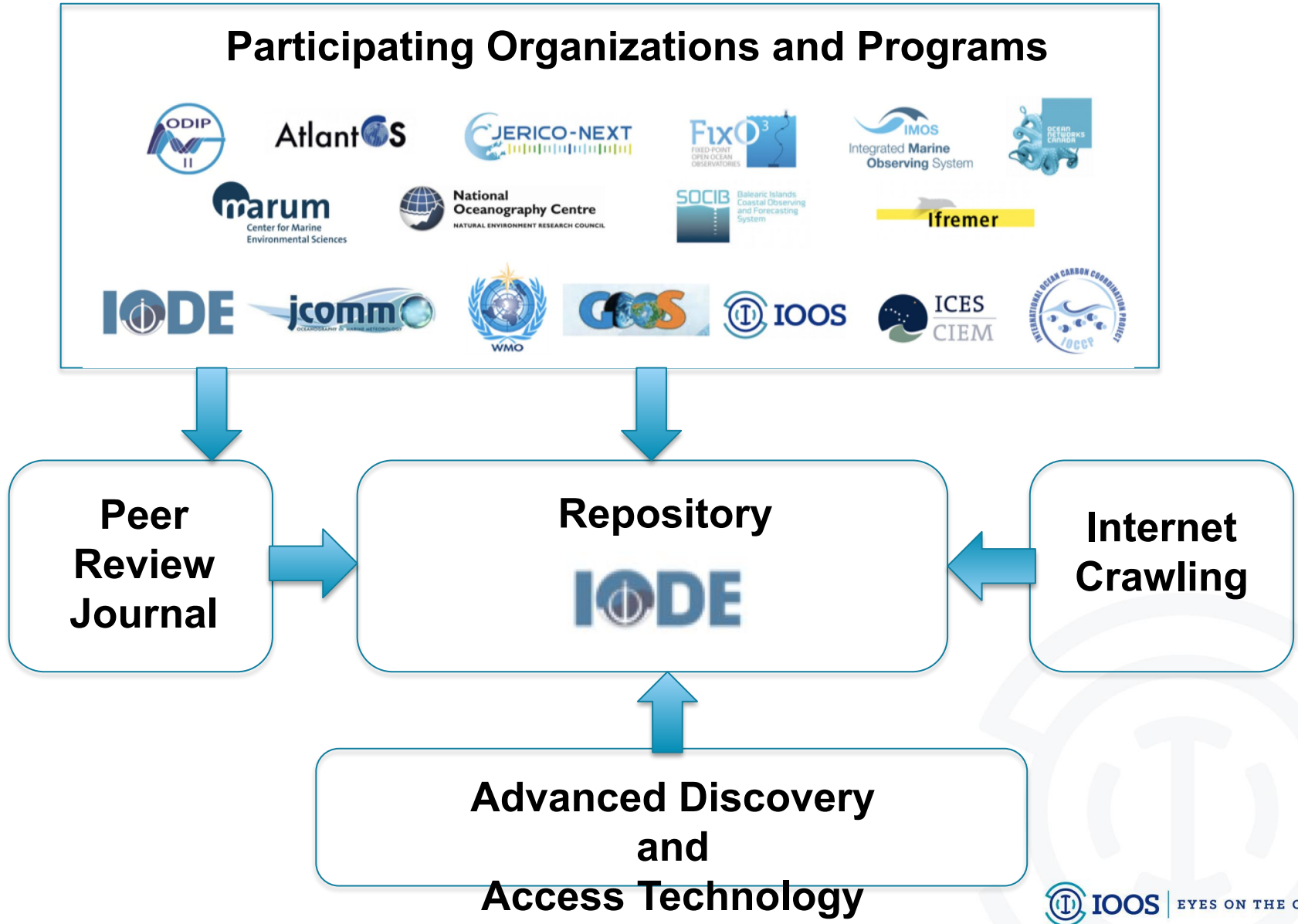
What is a Best Practice?

“A community best practice is a methodology that has repeatedly produced superior results relative to other methodologies with the same objective.

To be fully elevated to a best practice, a promising method needs to be adopted and employed by multiple organizations.”



Global Best Practice System



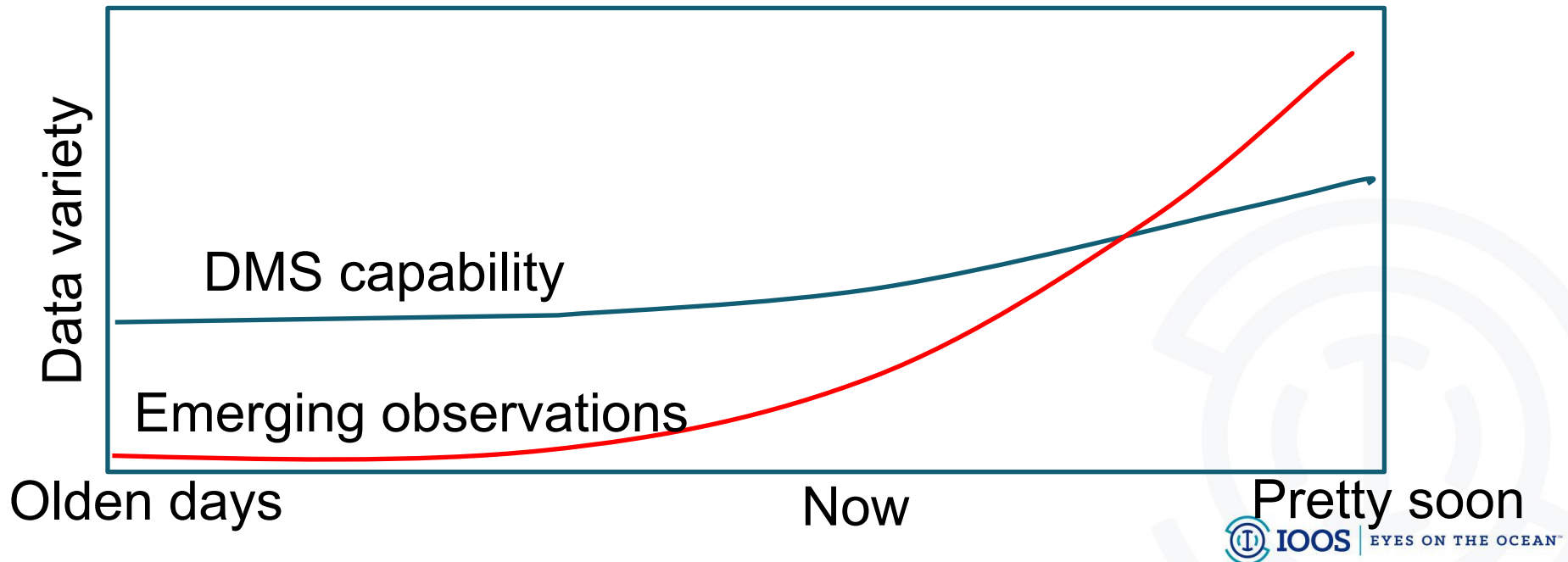
Best Practice System

- Expand IODE repository capabilities
 - Expand Permanent IDs through DOI, ORCHID
 - Implement Natural Language search drawing on marine vocabularies; compatibility with machine readability
 - Internet Crawling as inputs of additional practices
- Alignment with FAIR principles and open operations
- Introduction of community peer review including Research Topic in Frontiers in Marine Science

Future Data Management Services RT QC – Thoughts?



- Technology is making it cheaper & easier for more observers to obtain new types of data
- Existing DMS can't absorb it – more specialized DMS entities emerge
- Decentralization is a challenge, but QC closer to data source is a good thing
- Big DMS must guide emerging DMS standards & requirements



Summary

- **QARTOD is broadly accepted, and implementation partnerships might be more widely explored.**
- **Ocean Best Practices is worthy of your consideration.**
- **How can we prepare ourselves for the next decade of RT QC?**

Thanks!